Lesson 1 of 5

A square that moves

Make a window and steer a square with the arrow keys.

Goal	A window with a square you can steer with the arrow keys.
Concepts	import, variables, creating objects, functions, key events.
Students need	Python 3 with IDLE (or any editor) and the built-in turtle module.
Teacher prep	Have the finished code ready to run on the projector.

Session plan (30 minutes)

Time	Activity
0-5	Hook: run the finished Snake from Lesson 5 so students see where the course is heading. Today's piece is only the moving square.
5-15	Live-code together: window, square, one function, then the key bindings.
15-25	Students type the full program and test every arrow key.
25-30	Challenge and wrap-up: share who changed what.

The code

Students type this in full. Typos are where the learning happens.

```
import turtle

STEP = 20

screen = turtle.Screen()
screen.title("Snake")
screen.bgcolor("black")
screen.setup(600, 600)

head = turtle.Turtle("square")
head.color("lime")
head.penup()

def go_up():
    head.sety(head.ycor() + STEP)

def go_down():
    head.sety(head.ycor() - STEP)

def go_left():
    head.setx(head.xcor() - STEP)

def go_right():
    head.setx(head.xcor() + STEP)

screen.listen()
screen.onkeypress(go_up, "Up")
screen.onkeypress(go_down, "Down")
screen.onkeypress(go_left, "Left")
screen.onkeypress(go_right, "Right")

screen.mainloop()
```

How it works, in plain words

- **import turtle** loads a toolbox of drawing and game commands.
- **STEP = 20** is a named number, so we can change the speed in one place.
- **turtle.Turtle("square")** creates an object, our snake's head.
- **penup()** lifts the pen so the square does not draw a line behind it.
- A **function** (def) is a named set of instructions. Each arrow key runs one.
- **screen.onkeypress(go_up, "Up")** means: when Up is pressed, run go_up.
- **mainloop()** keeps the window open and listening.

Teaching notes

> Run the program **without** penup() first, so students see the square draw lines, then add it. Ask: what would happen if STEP were 50? Try it. If a key does nothing, click on the turtle window once so it has focus.

Challenge

- Change the colour and shape (try "circle" or "triangle").
- Make the square move faster or slower by changing STEP.
- Stretch: change the background colour of the window.

Common problems

What you see	Likely cause and fix
Window opens, keys do nothing	Click the turtle window first. Check that screen.listen() is there.
IndentationError	Lines inside a def must be indented by four spaces.
NameError: name 'turtl' is not defined	A typo. Read the name in the error and compare it with the code.
Window closes straight away	screen.mainloop() is missing at the end.

What we learned (student checklist)

[ ] I can import a module.

[ ] I can make a turtle object and change how it looks.

[ ] I can write a function with def.

[ ] I can connect a key press to a function.